

QUESTIONS 8 / 8 completed

Logger

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 interface Logger{
3     void debug(String m);
4     void info(String m);
5     void warning(String m);
6     void error(String m);
7 }
8 class lologger implements Logger{
9     public void debug(String m){System.out.println("DEBUG: "+m);}
10    public void info(String m){System.out.println("INFO: "+m);}
11    public void warning(String m){System.out.println("WARNING: "+m);}
12    public void error(String m){System.out.println("ERROR: "+m);}
13 }
14 }
```

OUTPUT

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Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes

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Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Character hero = new Hero("Hero");
4         Character villain = new Villain("Villain");
5         Character monster = new Monster("Monster");
6
7         hero.attack(monster);
8         villain.attack(hero);
9         monster.attack(villain);
10    }
11 }
12 interface Character {
13     void attack(Character target);
14     void takeDamage(int dmg);
15     void display();
16 }
17 abstract class BaseCharacter implements Character {
18     String name;
19     int health;
20     BaseCharacter(String name, int health) {
21         this.name = name;
```

OUTPUT

Hero attacked!

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Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

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Java

Run

Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         List<Integer> list = new ArrayList<>();
5         list.add(5);
6         list.add(2);
7         list.add(9);
8         list.add(1);
9         Collections.sort(list);
10        System.out.println(list);
11    }
12 }
```

OUTPUT

```
[1, 2, 5, 9]
```

QUESTIONS 8 / 8 completed

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.HashSet;
2 public class Main{
3     public static void main(String[] args){
4         Set<String> set = new HashSet<>();
5         set.add("Apple");
6         set.add("Banana");
7         set.add("Cherry");
8         set.add("Date");
9         System.out.println("HashSet:"+set);
10        set.remove("Banana");
11        System.out.println("HashSet after removal:"+set);
12    }
13 }
```

OUTPUT

Security Error: Disallowed code pattern detected.

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TreeMap

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Map;
2 import java.util.TreeMap;
3 public class Main{
4     public static void main(String[] args){
5         TreeMap<String,Integer>map=new TreeMap<>();
6         map.put("Apple",10);
7         map.put("Banana",25);
8         map.put("Date",30);
9         map.put("Cherry",15);
10        for(Map.Entry<String,Integer>entry:map.entrySet()){
11            System.out.println(entry.getKey()+":"+entry.getValue());
12        }
13    }
14 }
```

OUTPUT

```
Apple:10
```

QUESTIONS 8 / 8 completed

PriorityQueue

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.PriorityQueue;
2 public class Main{
3     public static void main(String[] args){
4         PriorityQueue<Integer> pq=new PriorityQueue<>();
5         pq.add(10);
6         pq.add(20);
7         pq.add(15);
8         pq.add(5);
9         System.out.println("PriorityQueue:"+pq);
10    }
11 }
```

OUTPUT

QUESTIONS 8 / 8 completed

LinkedListMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.LinkedHashMap;  
2 import java.util.Map;  
3 public class Main{  
4     public static void main(String[] args){  
5         LinkedHashMap<String,Integer>map=new LinkedHashMap<>();  
6         map.put("Apple",10);  
7         map.put("Banana",25);  
8         map.put("Cherry",15);  
9         map.put("Date",30);  
10        System.out.print("Keys in insertion order:");  
11        for(String key: map.keySet()){  
12            System.out.println(key);  
13        }  
14    }  
15 }
```

OUTPUT

```
Keys in insertion order:Apple
```

QUESTIONS 6 / 8 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         int[] a = {5, 2, 9, 1, 7};
5         int min = a[0], max = a[0];
6         for (int i = 1; i < a.length; i++) {
7             if (a[i] < min) min = a[i];
8             if (a[i] > max) max = a[i];
9         }
10        System.out.println("Min = " + min);
11        System.out.println("Max = " + max);
12    }
13 }
```

OUTPUT